SECTION 00 00 10

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PART 1 GENERAL

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END OF SECTION

SECTION 01 10 00
SUMMARY OF WORK

PART 1 GENERAL

1.1 WORK COVERED BY CONTRACT DOCUMENTS

A. This Section summarizes the Work to be performed under this Contract. The Contractor shall provide all labor, materials, equipment, and services necessary for the complete execution of the reroofing project, as indicated or reasonably implied by the Contract Documents.

1.2 RE-ROOFING WORK

A. This project includes the following reroofing work:

1. Windsor High School:

a. <u>BASE BID - All Sections</u>

i. Existing - Sloped metal deck, 6” Polyiso, Crickets in locations, Ballasted EPDM.

ii. New Installation:

1. Ballast Relocation: Carefully move the existing rock ballast in workable sections to expose the underlying roof system.
2. Ballast Cleaning: Screen and sieve the existing rock ballast to remove dirt, debris, and broken, sharp fragments to protect the new membrane.
3. Completely remove and dispose of the existing EPDM membrane and the underlying 1/2" wood fiberboard.
4. Loose Lay additional new 1" insulation board (20 psi).
5. Install new 1/2" fire-rated plywood sheathing to all parapet walls and equipment curbs.
6. Loose-lay a new 90-mil EPDM membrane over the insulation.
7. Install a manufacturer-approved membrane protection mat directly over the new 90-mil EPDM
8. Carefully reinstall the cleaned and sieved rock ballast over the protection mat. Distribute ballast per pattern and weight.
9. New edge metal, scuppers, conductor heads and downspouts.
10. Install new gas line supports at all gas & conduit lines, Paint all gas lines.
11. Disconnect / reconnect all heat tape lines in downspouts.

iii. Manufacturer Warranty - 30yr / 72mph / 2” hail / 32hr accidental puncture / metal edge to edge.

iv. 2 years from substantial completion.

b. Alternate 1 - All Sections
    i. New Installation:
        1. Remove Ballast and dump at a location in the district.
        2. Remove EPDM Membrane.
        3. Leave existing Polyiso insulation.
        4. Mechanically attach a new 1/2" cover board.
        5. Install new 1/2" fire-rated plywood sheathing to all parapet walls and equipment curbs.
        6. Fully adhere the new 145-mil Fleeceback EPDM membrane.
        7. New edge metal, scuppers, conductor heads and downspouts.
        8. Install new gas line supports at all gas & conduit lines, Paint all gas lines.
        9. Disconnect reconnect all heat tape lines in downspouts.
    ii. Manufacturer Warranty - 30yr / 90mph / 3” hail / 32hr accidental puncture / Metal (edge to edge).
    iii. 2 years from substantial completion.

1.4 PERMITS

A. The contractor shall be responsible for obtaining, paying for, and submitting all required building permits, including the primary roofing permit and any ancillary mechanical, electrical, or plumbing (MEP) permits required by state or local codes. The Consultant shall coordinate with the Contractor to provide necessary design documents and technical specifications to facilitate the submission to the Permitting Authority

PART 2 NOT USED

PART 3 NOT USED

END OF SECTION 01 10 00

SECTION 01 20 00
PRICE AND PAYMENT PROCEDURES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Price and payment procedures, including:
   1. Alternates
   2. Allowances
   3. Payment Applications

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION

3.1 ALTERNATES

A. Total Price:
   1. Provide the total price for each Alternate in the Bid Form.
   2. Include the cost of modifications to other work to accommodate the Alternate, along with related costs such as overhead and profit.

B. Acceptance of Alternates:
   1. The Owner will determine which Alternates, if any, are selected for inclusion in the Contract.

C. Coordination of Alternates:
   1. Modify or adjust affected adjacent work as necessary to integrate the work of the selected Alternate(s) into the Project.
   2. Coordinate Alternates with related work to ensure proper completion.

D. List of Alternates: NA

3.2 ALLOWANCES

A. General:
   1. Lump sum and unit cost allowances are specified below and as indicated on the Drawings.
   2. Amounts shall include all costs, including overhead and profit, except where noted.
   3. Coordinate allowances with requirements for related and adjacent work.

B. Unit Cost Allowances:
   1. The following items will be selected at a later date, and the unit costs below are for materials only. Installation and all other associated costs are to be included in the base bid. Refer to the Drawings to determine the required quantities, multiply by the unit cost, and include the total within the base bid.

a. Wood Decking Replacement: $___/sqft
b. Wood Blocking Replacement: $___/board foot
c. Metal Decking Replacement: $___/sqft

3.3 CONTRACT MODIFICATIONS

A. When modifications to the Contract are required, standard AIA Contract Documents shall be used to establish the details of the modifications and execute the change.

3.4 APPLICATION FOR PAYMENT

A. Submit Applications for Payment in accordance with one of the following options:
1. AIA Form G702: Application and Certificate for Payment.
2. Contractor's Standard Invoice: Include the following information at a minimum:
a. Original contract amount.
b. Any approved Change Orders and the current adjusted contract amount.
c. Percentage of work complete or value of materials stored on-site.

END OF SECTION

SECTION 01 30 00
ADMINISTRATIVE REQUIREMENTS

PART 1 GENERAL

1.1 SUMMARY

A. This Section includes requirements for project coordination, supervision, meetings, schedules, and photographic documentation for the reroofing project.

1.2 COORDINATION

A. Project Coordination:

This project requires careful coordination due to:

1. Complex nature and phasing of work activities.
    a. Occupancy of the facility by the Owner during construction.
2. The Contractor shall provide supervision, planning, scheduling, and control to meet Contract Documents requirements.
3. Coordinate construction operations in the required sequence for best results.
4. Notify affected parties in writing of any special coordination procedures.
5. Coordinate administrative procedures, including:
    a. Construction schedule and Schedule of Values.
    b. Temporary facilities and controls.
    c. Submittal processing.
    d. Progress and pre-installation meetings.
    e. Project closeout activities.
6. Notify the Owner's Representative in writing 48 hours in advance when returning work areas to the Owner or requiring new work areas.
7. Submit a building access plan to the Owner's Representative for approval at least 10 working days prior to implementation.

1.3 SUPERVISION

A. Project Manager:
    1. Provide a project manager on-site weekly for a weekly progress meeting during the Work.
    2. The superintendent shall be literate and fluent in English.

B. Superintendent
    1. Provide a project superintendent on-site for a minimum of four hours per week during the Work.
    2. The superintendent shall be literate and fluent in English.

C. Forman:
    1. Provide a project foreman on-site for a minimum of eight hours per day during the Work.
    2. The foreman shall be literate and fluent in English.

1.4 MEETINGS

A. General:
   1. Schedule and conduct meetings at the Site unless otherwise indicated.
   2. Notify participants of meeting dates and times.
   3. Prepare and distribute meeting agendas.
   4. The Consultant will record minutes and distribute them to concerned parties within seven days.

B. Pre-Construction Meeting:
   1. Conduct a pre-construction meeting before Work begins. Attendees shall include the Owner's Representative, Consultant, major subcontractors, and other concerned parties.
   2. Discuss the construction schedule, phasing, and critical work sequencing.
   3. Discuss the following:
      a. Subcontractor responsibilities and personnel assignments.
      b. Key personnel and contact information.
      c. Procedures for requests for interpretations, field decisions, and Change Orders.
      d. Procedures for processing Applications for Payment.
      e. Use of premises, including office and storage areas, parking, and Owner's requirements.
      f. Work hours and restrictions.
      g. Deliveries and priorities.
      h. Temporary facilities and controls.
      i. Housekeeping procedures, including waste management and recycling.
      j. Preparation of record documents.

C. Progress Meetings:
   1. Conduct progress meetings at regular intervals with the Owner's Representative, Consultant, contractors, subcontractors, suppliers, and other concerned parties.
   2. Review and approve previous meeting minutes.
   3. Review the construction schedule, identify any delays, and discuss corrective actions.
   4. Review the schedule for the next period.
   5. Discuss the following:
      a. Sequence of operations, interface requirements, and coordination.
      b. Status of submittals, deliveries, and off-site fabrication.
      c. Field observations, problems, and decisions.
      d. Quality and work standards
      e. Status of payment requests, Change Orders, and claims.

1.5 SCHEDULES

A. Construction Schedule:

1. Prepare a construction schedule for the entire Work, including a complete sequence of activities.
2. Use a horizontal bar chart format.
3. Provide beginning and completion dates for each activity and phase.
4. Include:
    a. Monthly completion percentages for each activity.
    b. Time periods when portions of the Site are unavailable for Owner use.
    c. Periods of utility service interruptions.
    d. Submittal dates.
    e. Product procurement and delivery dates.
    f. Dates for the selection of finishes.
5. Provide separate sub-schedules for critical portions of the Work.

B. Schedule Submission and Updates:
1. Submit the construction schedule within one week after the Notice to Proceed.
2. Update the schedule monthly or when actual progress deviates significantly.
3. Include a narrative report with the updated schedule discussing changes, problem areas, and corrective actions.
4. Distribute the current schedule to the job-site file, subcontractors, and other affected parties.

1.6 PHOTOGRAPHIC DOCUMENTATION

A. Photograph existing conditions, concealed conditions, finish surfaces, and construction damage.
B. Submit digital images to the Consultant within two days.
C. Submit photographs of pre-existing damage before starting work.

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION (Not Used)

END OF SECTION 01 30 00

SECTION 01 70 20
PROJECT CLOSEOUT

PART 1 GENERAL

1.1 SUMMARY

A. This Section includes administrative and procedural requirements for contract closeout, including:
   1. Final cleaning.
   2. Substantial Completion and final completion procedures.
   3. Project record documents.

1.2 SUBMITTALS

A. Warranties:
   1. Organize warranty documents in an orderly sequence based on the Project Manual's table of contents.
   2. Create into a single indexed electronic PDF file with a table of contents and navigational links.
   3. Submit one set of bound warranties and one scanned copy.

B. Product Maintenance Manual:
   1. Assemble a complete set of maintenance data for each product, material, and finish, including:
      a. Manufacturer's information, cross-referenced to Specification sections.
      b. Installer/supplier contact information.
      c. Maintenance schedules and procedures.
      d. Maintenance materials and sources.
      e. Maintenance record forms.
      f. Copies of warranties and maintenance agreements.
   2. Organize into sets with a separate section for each product.
   3. Bind and index data in heavy-duty, three-ring, vinyl-covered, loose-leaf binders.
   4. Provide dividers with tabs for each product.
   5. Label binders with "PRODUCT MAINTENANCE MANUAL," Project name, and subject matter.
   6. Scan the maintenance manual and assemble into a single indexed electronic PDF file with a table of contents and navigational links.
   7. Submit one set of bound manuals and one scanned copy.

PART 2 PRODUCTS

2.1 CLEANING AGENTS

A. Use cleaning materials and agents recommended by the manufacturer of the surface to be cleaned.
B. Do not use agents that are hazardous or could damage finished surfaces.

PART 3 EXECUTION

3.1 FINAL CLEANING

A. General: Clean the site to comply with local and federal regulations. Restore surfaces to their pre-construction condition.
    1. Remove tools, equipment, and surplus material.
    2. Clean the site of rubbish and waste materials.
    3. Clean exposed finishes to a dirt-free condition. Touch up or replace damaged finishes.
    4. Clean and restore transparent surfaces.
    5. Remove non-permanent labels.
    6. Clean debris from limited access spaces.
    7. Clean floors and carpets.
    8. Wipe down equipment surfaces.

3.2 SUBSTANTIAL COMPLETION

A. Before requesting inspection for Substantial Completion:
    1. Prepare a punch list with the value of incomplete items and reasons for incompleteness.
    2. Deliver tools, spare parts, and extra materials to the Owner.
    3. Remove temporary facilities, mockups, and construction tools.
    4. Complete final cleaning.
    5. Touch up and repair marred finishes.

3.3 FINAL COMPLETION

A. Before requesting final inspection:
    1. Submit the Consultant's Substantial Completion punch list, endorsed and dated, with a statement that items have been completed.
    2. Instruct Owner's personnel in product maintenance.
B. Request final inspection. The Consultant will inspect or notify the Contractor of unfulfilled requirements.

3.4 PROJECT RECORD DOCUMENTS

A. Maintain one set of Drawings, Shop Drawings, Specifications, and product data for recording as-built deviations.
    1. Accurately document changes and modifications.
    2. Record information on concealed elements.
    3. Include dimensional changes, revisions to details, changes by Change Order, field

records, and actual products used.

4. Mark the most relevant record document and cross-reference others.
5. Use red pencil for markups and other colors to distinguish categories of work.

B. Store Record Documents and samples separately from Contract Documents. Maintain them in good order and accessible to the Consultant.

END OF SECTION 01 70 20

SECTION 06 17 13
FIRE-RATED PLYWOOD SHEATHING AND WOOD NAILERS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Fire-rated plywood roof sheathing panels.

B. Wood nailers for attaching fire-rated plywood sheathing.

C. Fasteners for attaching nailers to structural supports and sheathing to walls.

1.2 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions, apply to this Section.

1.3 SUBMITTALS

A. Product Data: For each type of product indicated.

B. Shop Drawings: Show layout and installation details of fire-rated plywood sheathing and wood nailers, including:

1. Panel sizes and orientation
2. Nailer locations and sizes
3. Fastener types and spacing
4. Edge conditions and transitions

a. Fire-Retardant Treatment Certificate: From plywood manufacturer certifying compliance with specified fire-retardant standards.

1.4 QUALITY ASSURANCE

A. Fire-Retardant Treatment: Provide plywood and wood nailers with a fire-retardant treatment complying with ASTM E84, Class A.

B. Panel Grade: Provide plywood panels of the grade and span rating indicated on the Drawings.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials in manufacturer's original, unopened, undamaged containers with identification labels intact.

B. Store plywood panels flat in a dry, protected location, off the ground and supported to prevent sagging.

C. Protect panels from moisture and damage during handling and installation.

PART 2 PRODUCTS

2.1 FIRE-RATED PLYWOOD SHEATHING

A. Lumber, plywood, and Engineered Wood Products (EWP) bearing the PyroGuard™ mark has a flame spread rating of 25 or less (Class A) when tested in accordance with ASTM E84, “Standard Test Method for Surface Burning Characteristics of Building Materials.” PyroGuard™ fire-

retardant-treated wood shows no evidence of significant progressive combustion when the test is extended for an additional 20-minute period. In addition, the flame front does not progress more than 10½ feet beyond the centerline of the burners at any time during the test. The flame spread and smoke developed index for each species and product are classified by Underwriters Laboratories Inc. (UL).

B. PyroGuard™ fire-retardant-treated wood is manufactured under the independent third-party inspection of Underwriters Laboratories Inc. (UL) Follow-Up Service and each piece shall bear the UL classified mark indicating the extended 30 minute ASTM E84 test.

C. PyroGuard™ shall be kiln dried after treatment (KDAT). The kiln drying process is monitored by Underwriters Laboratories, Inc. (UL) and the UL mark shall appear on the label. D. PyroGuard™ shall be produced in accordance with UL Evaluation Report ER7002-01 (UL ER7002-01), latest version. E. PyroGuard™ meets the performance requirements of AWPA U1, Commodity Specification H for Use Category UCFA and AWPA C20/C27 (Type A, HT). F. PyroGuard™ is listed on the Department of Defense (DoD) Qualified Products List (QPL) and meets the requirements of MIL-L-19140E as a Type 1 fire-retardant treatment for lumber and plywood.

D. Use for exterior locations and where indicated on the drawings.

E. Thickness: 1/2 inch

2.2 WOOD NAILERS

A. Lumber, plywood, and Engineered Wood Products (EWP) bearing the PyroGuard™ mark has a flame spread rating of 25 or less (Class A) when tested in accordance with ASTM E84, “Standard Test Method for Surface Burning Characteristics of Building Materials.” PyroGuard™ fire-retardant-treated wood shows no evidence of significant progressive combustion when the test is extended for an additional 20-minute period. In addition, the flame front does not progress more than 10½ feet beyond the centerline of the burners at any time during the test. The flame spread and smoke developed index for each species and product are classified by Underwriters Laboratories Inc. (UL).

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D. Use for exterior locations and where indicated on the drawings.

E. Size: (2 to 4 inches thick, 2 to 4 inches wide) products to match existing size.

2.3 DIMENSION LUMBER

A. For light framing (2 to 4 inches thick, 2 to 4 inches wide) provide the following grade and species:
   1. Standard" grade.
   2. Any species graded under WWPA or WCLIB rules.
   3. Southern Pine graded under SPIB rules.

2.4 MISCELLANEOUS LUMBER

A. General: Provide treated lumber for support or attachment of other construction including rooftop equipment curbs and support bases, bucks, nailers, blocking, furring, grounds, stripping, and similar members. Fabricate miscellaneous lumber from dimension lumber of sizes indicated and into shapes shown.

B. Moisture content: 15 percent maximum for lumber items not specified to receive wood preservative treatment.

C. Grade: "Standard" grade light-framing-size lumber of any species or board-size lumber as required. "No. 3 Common" or "Standard" grade boards per WCLIB or WWPA rules or "No. 2 Boards" per SPIB rules.

2.5 FASTENERS

A. Type and Size: As recommended by the sheathing manufacturer for attachment to the specified nailers and structural supports.
   1. Wood / Metal - ¼” corrosion resistant fastener.
   2. Concrete / Masonry - ¼” corrosion resistant fastener.

PART 3 EXECUTION

3.1 INSTALLATION

A. INSTALLATION, GENERAL
   1. Discard units of material with defects that impair quality of rough carpentry construction and that are too small to use in fabricating rough carpentry with minimum joints.
   2. Set rough carpentry to required levels and lines, with members plumb and true to line and cut and fitted.
   3. Fit rough carpentry to other construction; scribe and cope as required for accurate fit. Correlate location of furring, nailers, blocking, grounds, and similar supports to allow attachment of other construction.
   4. Securely attach rough carpentry work to substrate by anchoring and fastening as indicated. Countersink all fasteners.
   5. Use common wire nails, unless otherwise indicated. Select fasteners of size that will not penetrate members where opposite side will be exposed to view or will receive finish materials. Make tight connections between members. Install fasteners without splitting of

wood; pre-drill as required.

6. Wood Blocking Installation:
    a. Refasten existing nailers to structure spaced at 6”o.c. staggered.
    b. Ensure nailers are aligned, level, and securely fastened.
    c. Install nailers perpendicular to roof joists or rafters.
    d. Attach new nailers to structural supports with ¼” corrosion resistant wood screws at 2 rows / 6” o.c. staggered.
    e. Ensure nailers are aligned, level, and securely fastened.
7. Sheathing Installation:
    a. Install fire-rated plywood sheathing at all designated locations.
    b. Fasten sheathing at 6” o.c. at edges and 12”o.c. in the field - 12” from the edge.

END OF SECTION

SECTION 07 01 50
PREPARATION FOR REROOFING

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes selective removal and off-site disposal of existing roofing components and other construction items as indicated on Drawings.

B. Existing roofing system descriptions are shown on the Drawings.

C. Protection of existing insulation and miscellaneous items to be reused is required.

1.3 MATERIALS OWNERSHIP

A. Demolished materials become Contractor's property and shall be removed from the Project site, except for items indicated to be reused, reinstalled, or remain Owner's property.

1.4 DEFINITIONS

A. Refer to ASTM D1079 and the glossary in NRCA's "The NRCA Roofing and Waterproofing Manual" for roofing terminology.

B. **Roof Tear-Off:** Complete removal of roofing system and insulation down to the structural deck.

C. **Partial Roof Tear-Off:** Removal of a portion of the existing membrane roofing system or selected components.

D. **Remove:** Detach items and legally dispose of them off-site unless indicated to be removed and reinstalled.

E. **Existing to Remain:** Existing items not indicated to be removed.

1.5 QUALITY ASSURANCE

A. Comply with EPA notification regulations before starting roof removal.

B. Comply with hauling and disposal regulations of authorities having jurisdiction.

1.6 PROJECT CONDITIONS

A. Occupancy:

1. Owner will occupy adjacent areas during demolition. Minimize disruption to Owner's operations.
2. Provide 72 hours' notice to Owner before demolition activities affecting operations.
3. Coordinate daily with Owner for protection of equipment, HVAC shutdown, and occupant evacuation.
4. Notify Owner to evacuate occupants before working over structurally impaired areas.

B. Condition of Structures:

1. Owner assumes no responsibility for the actual condition of items to be demolished.

2. Conditions at the time of bidding will be maintained insofar as practicable.

C. Protection:
   1. Protect the building, adjacent buildings, walkways, site improvements, plantings, and landscaping from damage.
   2. Provide temporary barricades and protection for personnel and the public.
   3. **Protect interior spaces from debris & clean any tear-off debris left in the ceiling.**
   4. Maintain safe passage for occupants and the public.
   5. Erect temporary covered passageways as required.
   6. Protect existing finishes to remain.
   7. Protect floors and seats with coverings during tear-off.
   8. Construct temporary dust-proof partitions as needed.
   9. Provide temporary weather protection to prevent water damage.
   10. Remove protections daily at the end of work.

D. Safety: Criminal Record Verification: Contractor will be required to complete Criminal Record Check on all employees who work on the Project. Employees who are not able to work in the USA legally and/or have been convicted of a felony, including but not limited to crimes involving physical assault, sexual assault, drug use, and crimes that require registration on the National Sexual Offender Registry will not be allowed to work on the project. The Owner reserves the right to disqualify other employees who have a felony conviction. The Contractor shall be responsible to the Owner for the acts and omissions of all his employees. The Contractor shall further be responsible for the acts and omissions of all subcontractors, their agents and employees, and all other persons acting on behalf or for the Contractor or subcontractors.

E. Access / Work Restrictions
   a. Driveways, walkways and entrances: Keep driveways and entrances serving premises clear and available to Owner, Owner's employees, and emergency vehicles at all times. Do not use these areas for parking or storage of materials.
   b. Coordinate all work with the building occupants, typically the school principal prior to beginning work.
   c. Smoking is not permitted on District property.

F. Roof Loads: Limit construction loads on the roof to 30 lbs. per square foot for uniformly distributed loads.

G. Weather Limitations: Proceed only when weather permits work without water entering the building.

H. Hazardous Materials:
   1. Hazardous materials are not expected but if encountered, do not disturb and notify Architect and Owner.
   2. Owner will remove hazardous materials under a separate contract.

3. Damages: Repair damages to adjacent facilities caused by demolition work.

I. Traffic: Minimize interference with roads, walks, and other facilities. Do not obstruct traffic without permission.

J. Flame Cutting: Do not use cutting torches until the area is clear of flammable materials. Verify concealed space conditions before flame cutting. Maintain fire suppression devices.

K. Utility Services: Maintain and protect existing utilities to remain in service.

L. Environmental Controls: Limit dust and dirt migration using water sprinkling, enclosures, etc. Comply with environmental regulations.

PART 2 PRODUCTS

2.1 TEMPORARY ROOFING MATERIALS

A. Design, material selection, and installation of temporary roofing are the Contractor's responsibility.

PART 3 EXECUTION

3.1 PREPARATION

A. Coordinate with Owner to shut down air-intake equipment and cover louvers.

B. Have sufficient materials on-site for temporary rain protection.

C. Maintain functioning roof drains and prevent debris from entering. Use plugs and remove them daily or when rain is forecast. Provide alternative drainage if drains are blocked.

D. Verify rooftop utilities are shut off before starting work.

E. Provide shoring and bracing to prevent movement or collapse of the demolition area and adjacent structures.

1. Verify existing conditions and notify the Consultant of discrepancies.
2. Cease operations and notify the Owner's Representative if structural safety is endangered.
3. Protect furniture, equipment, and fixtures from damage.
4. Disconnect and remove utility services not indicated to remain.

3.2 ROOF TEAR-OFF

A. General: Perform demolition systematically to protect adjacent construction.

1. Keep demolition to the minimum necessary.
2. Disconnect and shut off utilities before removing equipment.
3. Remove debris promptly to avoid overloading supporting structures.
4. Remove only the roofing that can be replaced the same day.
5. Use enclosed chutes or containers for debris removal. Do not throw materials from the roof.

B. Removal of Existing Roofing:

1. Remove the existing roofing system down to the structural deck.
2. Existing wood blocking and framing at parapet walls shall remain, except where indicated.
3. Prevent damage to structural decks and supports.
4. Remove existing curbs, mechanical equipment, and other items as indicated.

C. Unanticipated Conditions:

1. Investigate and report any unanticipated conflicts with mechanical, electrical, or structural elements.
2. Adjust the demolition schedule as needed to maintain progress.

3.3 DECK PREPARATION

A. Notify the Owner's Representative of any loose or damaged deck fasteners or if the deck is unsuitable for new roofing. Do not proceed until directed.

3.4 DISPOSAL

A. Collect and promptly dispose of demolished materials off-site. On-site storage or sale is not permitted.

END OF SECTION 07 01 50

SECTION 07 53 00

EPDM THERMOSET SINGLE-PLY ROOFING - *BASE BID*

PART 1 GENERAL

1.01 SECTION INCLUDES

A. EPDM thermoset single-ply roofing
B. Membrane flashings.
C. Metal flashings.
D. Roof insulation.

1.02 RELATED SECTIONS

A. Section 05 31 23 – Steel Roof Decking.
B. Section 06 10 00 – Rough Carpentry.
C. Section 07 62 00 – Sheet Metal Flashing and Trim.
D. Section 07 70 00 – Roof and Wall Specialties and Accessories.

1.03 REFERENCES

A. American Society of Civil Engineers (ASCE) – ASCE 7 - Minimum Design Loads for Buildings and Other Structures, Current Revision.
B. ANSI/SPRI WD-1 “Wind Design Standard for Roofing Assemblies”.
C. ASTM International (ASTM):
    1. ASTM C 1289 – Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board.
    2. ASTM D 412 – Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers-Tension.
    3. ASTM D 624 – Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers.
    4. ASTM D 816 – Standard Test Methods for Rubber Cements.
    5. ASTM D 4637 – Standard Specification for EPDM Sheet Used In Single-Ply Roof Membrane.
    6. ASTM E 96 – Standard Test Methods for Water Vapor Transmission of Materials.
D. International Code Council (ICC):
    1. International Building Code (IBC).
E. National Roofing Contractors Association (NRCA) – Low Slope Roofing and Waterproofing Manual, Current Edition.
F. Sheet Metal and Air Conditioning Contractors National Association, Inc. (SMACNA) – Architectural Sheet Metal Manual.
G. Underwriters Laboratories (UL):
    1. TGFU R1306 – “Roofing Systems and Materials Guide”

2. UL-790 – Standard Test Method for Fire Tests of Roof Coverings.

H. ANSI/ASHRAE/IESNA Standard 90.1 (2007): Energy Standard for Buildings Except Low-Rise Residential Buildings

1.04 DESIGN CRITERIA

A. Wind Uplift Performance:
   1. Roof system is designed to withstand wind uplift forces as calculated using the current revision of ASCE-7.
   2. Design Wind Speed - 115 ultimate wind speed.
   3. 72 MPH manufacturer wind speed warranty.

B. Fire Resistance Performance:
   1. Roof system will achieve a UL Class A rating when tested in accordance with UL-790.

C. Hail Rating: Severe Hail (SH)

D. Thermal Performance: Roof system will achieve a minimum R value not less than 30.

E. Drainage: Provide a roof system with positive drainage where all standing water dissipates within 48 hours after precipitation ends.

F. Building Codes:
   1. Roof system will meet the requirements of all federal, state and local code bodies having jurisdiction.

1.05 SUBMITTALS

A. Submit under provisions of Section 01 30 00.

B. Product Data: Manufacturer's data sheets on each product to be used, including:
   1. Preparation instructions and recommendations.
   2. Storage and handling requirements and recommendations.
   3. Installation methods.

C. Detail Drawings:
   1. Submit approved plan, section, elevation or isometric drawings which detail the appropriate methods for all flashing conditions found on the project.
   2. Coordinate approved drawings with locations found on the Contract Drawings.

D. Roof Assembly Letter: Provide a roof assembly letter on [Roof Manufacturer's] letterhead for each roof area, detailing the following:
   1. System Components:
      a. List all components of the roofing assembly in the order of installation, from the structural deck up to the membrane.
      b. Include the manufacturer's name and product name for each component.
   2. Fastening Patterns and Zones:
      a. Clearly indicate the fastening pattern for each component, including type of fastener, size, and spacing.

b. Delineate wind zones on the roof plan and provide corresponding fastening requirements for each zone based on ASCE 7 calculations.

3. Warranty Requirements:
   a. State the specific installation requirements to achieve the full manufacturer's warranty.
   b. Include any limitations or exclusions to the warranty.

E. Selection Samples: For each finish product specified, two complete sets of chips representing the manufacturer's full range of available colors, membranes, and thicknesses.

F. Verification Samples: For each finish product specified, two samples, minimum size 4 inches (100 mm) square representing actual product, color, and patterns.

1.06 QUALITY ASSURANCE

A. Manufacturer Qualifications: All products specified in this section will be supplied by a single manufacturer with a minimum of twenty (20) years experience.

B. Installer Qualifications:
   1. All products listed in this section are to be installed by a single installer with a minimum of five (5) years demonstrated experience in installing products of the same type and scope as specified.
   2. Installer must be capable of extending the Manufacturer's No Dollar Limit guarantee.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Store products in manufacturer's unopened packaging until ready for installation.

B. Store and dispose of hazardous materials, and materials contaminated by hazardous materials, in accordance with requirements of local authorities having jurisdiction.

C. Safety Data Sheets (SDS) must be on location at all times during the transportation, storage and application of materials.

D. When loading materials onto the roof, the Roofing Applicator must comply with the requirements of the building owner to prevent overloading and possible disturbance to the building structure.

1.08 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by the manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

B. When positioning membrane sheets, exercise care to locate all field splices away from low spots and out of drain sumps. All field splices should be shingled to prevent bucking of water.

C. Proceed with roofing work only when weather conditions are in compliance with the manufacturer's recommended limitations, and when conditions will permit the work to proceed in accordance with the manufacturer's requirements and recommendations.

D. Proceed with work so new roofing materials are not subject to construction traffic. When necessary, new roof sections shall be protected and inspected upon completion for possible damage.
E. Provide protection, such as ¾ inch thick plywood, for all roof areas exposed to traffic during construction. Plywood must be smooth and free of fasteners and splinters.
F. The surface on which the insulation or roofing membrane is to be applied shall be clean, smooth, dry, and free of projections or contaminants that would prevent proper application of or be incompatible with the new installation, such as fins, sharp edges, foreign materials, oil and grease.
G. New roofing shall be complete and weathertight at the end of the work day.
H. Contaminants such as grease, fats and oils shall not be allowed to come in direct contact with the roofing membrane.

1.09 WARRANTY

A. At project closeout, provide to the Owner or Owners Representative an executed copy of the manufacturer's Total-System warranty, outlining its terms, conditions, and exclusions from coverage.
    1. 30 years Total-System warranty, outlining its terms, conditions, and exclusions from coverage.
    2. Coverage to be extended to include damage caused by a maximum 72 mph wind speeds in accordance with terms stated in the Warranty document.
    3. Coverage to be extended to include damage caused by a maximum 2 inch diameter hail in accordance with terms stated in the Warranty document.
    4. Coverage to be extended to include 32 hours of accidental puncture warranty in accordance with the stated terms of the warranty document.
    5. Coverage to be extended to include roof edge metal water tightness in accordance with terms stated in the Warranty document.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Acceptable Manufacturer:
    1. Carlisle SynTec Systems (Basis of Design)
    2. Elevate Roofing Systems
    3. Johns Manville Roofing Systems
B. Requests for substitutions will not be considered.

2.02 SCOPE / APPLICATION

A. Roof System: Provide a waterproof roof system, capable of withstanding uplift forces as specified in Design Criteria.

B. Base Flashing: Provide a waterproof, fully adhered base flashing system at all penetrations, plane transitions and terminations.
C. Insulation: Provide a roof insulation system beneath the finish membrane.

2.03 ETHYLENE, PROPYLENE, DIENE TERPOLYMER (EPDM) MEMBRANE

A. Sure-Seal Non-Reinforced Membrane: Cured, non-reinforced EPDM membrane meeting the requirements of ASTM D 4637 Type I.
    1. Color: Black.
    2. Attachment Method: BallastedFully adhered.
    3. Membrane Thickness: 90 mil nominal.
    4. Sheet Dimensions:
        a. Width: 10 feet (3.05 m) maximum.
        b. Length: 100 feet (30.5 m) maximum.
    4. Performance:
        a. Tensile Strength: 1550 psi (10.7 MPa) minimum.
        b. Tear Resistance: 200 lbf/in (35 Kn/m) minimum.
            c. Elongation: 480 percent.

2.04 INSULATION

A. Polyisocyanurate InsulBase: Carlisle InsulBase. Rigid board with glass fiber reinforced facers (GRF) on both sides, meeting or exceeding the requirements of ASTM C 1289, Type II, Class 1.
    1. Compressive Strength: Grade 2 (20 psi) (138 kPa).
    2. Density: 2 lb per cubic foot (24 kg/cu m) minimum.
    3. Size: 1” thick

2.05 FLASHING ACCESSORIES

A. Sure-Seal (black)/Sure-White (white) Pressure-Sensitive Pipe Seals with Factory-Applied TAPE on the deck flange are available for use with Sure-Seal/Sure-White Roofing systems.
B. Sure-Seal/Sure-White Pressure-Sensitive (PS) Inside/Outside Corner: A 7 inch by 9 inch precut 60-mil thick Elastoform Flashing with a 30-mil Factory-Applied TAPE.
C. Sure-Seal/Sure-White Pressure-Sensitive Cured Cover Strip: Sure-Seal or Sure-White 60-mil cured EPDM membrane laminated to a nominal 35-mil cured Factory-Applied TAPE.
D. Sure-Seal/Sure-White Pressure-Sensitive “T” Joint Covers: A factory cut uncured 60-mil thick EPDM flashing laminated to a nominal 35-mil Factory-Applied TAPE, used to overlay field splice intersections and to cover field splices at angle changes. Available in 6 inch by 6 inch and 12 inch by 12 inch for Sure-Seal applications, and 6 inch by 6 inch for Sure-White applications.
E. Sure-Seal/Sure-White Pressure-Sensitive Elastoform Flashing: 60-mil thick uncured EPDM Flashing laminated to a 30-mil Factory-Applied Pressure-Sensitive TAPE used in conjunction with Sure-Seal Primer.
F. Sure-Seal Pressure-Sensitive RUSS (Reinforced Universal Securement Strip):

1. 6 inch (152 mm) RUSS: A nominal 6 inch (152 mm) wide, 45-mil thick reinforced EPDM membrane with a nominal 3 inch (76 mm) wide 30-mil thick cured synthetic rubber pressure-sensitive adhesive laminated to one edge. This product provides perimeter securement, and additional membrane securement at angle changes for Adhered, Ballasted, and Mechanically Fastened Roofing Systems.

2.06 CLEANERS, PRIMERS, ADHESIVES AND SEALANTS

A. Carlisle Weathered Membrane Cleaner: Clear, solvent-based cleaner used to loosen and remove contaminants from the surface of exposed EPDM membrane prior to applying EPDM Primer.

B. Sure-Seal SecurTAPE: 3 inch (76 mm) or 6 inch (152 mm) wide by 100 foot (30.5 M) long splice tape used for splicing adjoining sections of EPDM membrane.

C. Sure-Seal HP-250 Primer: A solvent-based primer used to prepare the surface of EPDM membrane for application of Splice Tape or Pressure-Sensitive products.

D. Sure-Seal/Sure-White Lap Sealant: A heavy-bodied material (trowel or gun-consistency) used to seal the exposed edges of a membrane splice.
   1. Sure-Seal Lap Sealant: Black sealant for use with Sure-Seal (black) Roofing Systems.
   2. Sure-White Lap Sealant: White sealant for use with Sure-White (white-on-black) Roofing Systems.

E. 90-8-30A Bonding Adhesive: A high-strength, yellow colored, synthetic rubber adhesive used for bonding Sure-Seal/Sure-White EPDM membranes to various surfaces.

F. Flexible FAST Adhesive: A spray or extruded applied, two-component, polyurethane, low-rise expanding foam adhesive used to securely bond FleeceBACK membranes to a variety of substrates.

G. Water Cut-Off Mastic: A one-component, low viscosity, self wetting, Butyl blend mastic used as a compression sealing agent between EPDM membranes and applicable substrates.

H. Universal Single-Ply Sealant: A 100 percent solids, solvent free, one-part, polyether sealant that provides a weather tight sealant to a variety of building substrates; used as a termination bar sealant. Available in white only.

2.07 EDGINGS AND TERMINATIONS

A. SecurEdge EX Snap-On Fascia: A snap-on edge system consisting of a 24 gauge galvanized metal water dam. Finish as noted on the Finish Schedule of the Contract Drawings.

2.8 FASTENING COMPONENTS

A. HP Fastener: Threaded, black epoxy electro-deposition coated (E-Coat) fastener for use with steel, wood plank or oriented strand board (OSB).

B. Seam Fastening Plate: 2 inch (51 mm) diameter metal plate for insulation, membrane and RUSS attachment.

C. Insulation Fastening Plate: Nominal 3 inch (76 mm) diameter FM approved metal plate used for insulation attachment.

2.9 PROTECTION MAT

A. A protective mat/cushion layer shall be installed directly over the 90-mil EPDM membrane prior to ballast placement to prevent puncture and mechanical damage. BALLAST

2.10 BALLAST

B. 90-mil EPDM membrane loose laid and held in place with 1-1/2" nominal rounded water worn gravel conforming to gradation #4 in accordance with the ASTM-D448 method of sizing. Coverage rate shall be a minimum of 12 pounds per square foot in the field and 2-1/2” nominal conforming to graduation #1 or #2 in zone 3 at a coverage rate of 15 pounds per square foot in minimum 12’ x 12’ corners.

2.11 CONCRETE ROOF WALKWAY PAVER

A. Furnish and install Westile High-Density Concrete Roof Pavers (24" x 24" x 2" nominal, minimum 8,000 PSI) featuring chamfered edges and a standard slip-resistant finish, placed directly over a manufacturer-approved 60mil EPDM membrane protection pad to establish a designated maintenance walkway.

PART 3 EXECUTION

3.01 EXAMINATION

A. Do not begin installation until substrates have been properly prepared.

B. If substrate preparation is the responsibility of another installer, notify the Consultant of unsatisfactory preparation before proceeding.

3.02 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

C. Do not commence work until all other work trades have completed jobs that require them to traverse the deck on foot or with equipment.

3.025 BALLAST REMOVAL AND PREPARATION

A. Equipment Construction (Box Screen): The Contractor shall clean the existing rock ballast utilizing a site-built or manufactured "Box Screen." This apparatus must consist of a rigid framed box with heavy-duty wire mesh stretched securely across the top or middle, positioned over a debris catch basin or containment area.

B. B. Execution & Sifting Process: Workers shall shovel the existing dirty ballast directly onto the wire mesh and manually agitate the material to ensure all dirt, pine needles, and broken, sharp rock fragments (fines) fall through the mesh into the catch below.

C. C. Quality Control & Debris Removal: Only smooth, clean, full-sized ballast rock retained on top of the mesh shall be approved for reinstallation over the new roof assembly. The sifted waste material captured in the lower box must be regularly emptied, bagged, and legally disposed of

off-site.

D. D. Membrane & Deck Protection: To prevent puncture damage or crushing of the insulation during the shoveling and sifting process, the box screen setup must be placed over protective plywood or heavy-duty protection mats at all times while in operation.

3.03 INSULATION – SYSTEM DESIGN

A. Base Layers:
   1. Type: Polyisocyanurate 20 PSI – ASTM C 1289, Type II, Class 1.
   2. Thickness: 1”.
   3. Attachment Method: Clean & blow off existing insulation - Loose Laid

3.04 INSULATION PLACEMENT

A. Clean off existing insulation to remain.
B. Install insulation or membrane underlayment in multiple layers over the substrate with boards butted tightly together with no joints or gaps greater than ¼ inch (6 mm). Stagger joints both horizontally and vertically.
C. Secure insulation to the substrate with the required mechanical fasteners or insulation adhesive in accordance with the manufacturer’s current application guidelines.
D. Do not install wet, damaged or warped insulation boards.
E. Stagger joints in one direction unless joints are to be taped. Install insulation boards snug. Gaps between board joints shall not exceed ¼ inch (6 mm). Fill all gaps in excess of ¼ inch (6 mm) with same insulation material.
F. Wood nailers must be at least 3 ½ inches (89 mm) wide or 1 inch (25 mm) wider than adjacent metal flanges. Thickness must equal that of insulation but not less than 1 inch (25 mm) thickness.
G. Miter and fill the edges of the insulation boards at ridges, valleys and other changes in plane to prevent open joints or irregular surfaces. Avoid breaking or crushing the insulation at the corners.
H. Do not install any more insulation than will be completely waterproofed each day.

3.05 INSULATION ATTACHMENT

A. Attachment: Insulation boards shall be laid loose without mechanical fasteners or adhesive, relying on the overlying ballasted system assembly for securement.

3.06 MEMBRANE PLACEMENT AND ATTACHMENT (Ballasted System)

A. Position and unroll membrane sheets over the prepared substrate without stretching. Allow sheets to relax unconstrained for a minimum of 30 minutes prior to splicing or securing.
B. Align adjacent sheets to provide for a minimum 3-inch (76 mm) side lap and 6-inch (152 mm) end lap, or per manufacturer guidelines for ballast-secured splice details.
C. Membrane sheets shall be laid loose over the substrate without adhesive application, except at perimeters, penetrations, wall flashings, and angle changes where mechanical attachment or full adhesion is required per specification and manufacturer details.

D. Complete all field splices, end laps, and perimeter securement in accordance with manufacturer technical specifications prior to the placement of protective layers or ballast.
E. Protection Mat Installation: Unroll and lay a non-woven polypropylene geotextile protection mat over the completed, inspected membrane prior to ballast placement. Overlap all adjacent protection mat sheets a minimum of 6 inches (152 mm) to ensure continuous protection against ballast abrasion and puncture.

3.07 MEMBRANE SPLICING (Tape Splice – Factory Applied)

A. Overlap adjacent sheets and mark a line ½ inch out from the top sheet.
B. Fold the top sheet back and clean the dry splice area (minimum 2 ½ inches (64 mm wide) of both membrane sheets with Sure-Seal Primer as required by the membrane manufacturer.
C. Where Splice Tape is not Factory-Applied, apply Splice Tape to bottom sheet with the edge of the release film along the marked line. Press tape onto the sheet using hand pressure. Overlap tape roll ends a minimum of 1 inch (13 mm).
D. Remove the release film and press the top sheet onto the tape using hand pressure.
E. Roll the seam toward the splice edge with a 2 inch (51 mm) wide steel roller.
F. Install Pressure-Sensitive “T” Joint Cover, a 6 inch wide (152 mm) section of Pressure-Sensitive Elastoform Flashing over all field splice intersections.
G. When using non-Pressure-Sensitive Elastoform Flashing or Elastoform Flashing, seal edges of flashing with Lap Sealant.
H. The use of Lap Sealant with tape splices is optional except at tape overlaps and cut edges of reinforced membrane where Lap Sealant is required.

3.08 FLASHING

A. Wall and curb flashing shall be 90 MIL cured EPDM membrane. Continue the deck membrane as wall flashing where practicable.
B. Follow manufacturer’s typical flashing procedures for all wall, curb, and penetration flashing including metal edging/coping and roof drain applications.

3.09 WALKWAYS

A. Install walkways at all traffic concentration points (such as roof hatches, access doors, rooftop ladders, etc.) and all locations as identified on the Contract Drawings.
B. Non-ballasted sections - Adhere walkways pads to the EPDM membrane in accordance with the manufacturer’s current application guidelines.
C. Ballasted sections - loose lay 60mil EPDM membrane under new concrete paver in accordance with the manufacturer’s current application guidelines.

3.10 DAILY SEALS

A. On phased roofing, when the completion of flashings and terminations is not achieved by the end of the work day, a daily seal must be performed to temporarily close the membrane to prevent water infiltration.

B. Use Sure-Seal Pourable Sealer or other acceptable membrane seal in accordance with the manufacturer's requirements.

3.11 CLEAN UP

A. Perform daily clean-up to collect all wrappings, empty containers, paper, and other debris from the project site. Upon completion, all debris must be disposed of in a legally acceptable manner.

B. Prior to the manufacturer's inspection for warranty, the applicator must perform a pre-inspection to review all work and to verify all flashing has been completed as well as the application of all caulking.

3.12 PROTECTION

A. Protect installed products until completion of project.

B. Touch-up, repair or replace damaged products before Substantial Completion.

END OF SECTIONSECTION 07 53 00

SECTION 07 53 00

EPDM THERMOSET SINGLE-PLY ROOFING - *ALTERNATE 1*

PART 1 GENERAL

1.01 SECTION INCLUDES

A. EPDM thermoset single-ply roofing
B. Membrane flashings.
C. Metal flashings.
D. Roof insulation.

1.02 RELATED SECTIONS

A. Section 05 31 23 – Steel Roof Decking.
B. Section 06 10 00 – Rough Carpentry.
C. Section 07 62 00 – Sheet Metal Flashing and Trim.
D. Section 07 70 00 – Roof and Wall Specialties and Accessories.

1.03 REFERENCES

A. American Society of Civil Engineers (ASCE) – ASCE 7 - Minimum Design Loads for Buildings and Other Structures, Current Revision.
B. ANSI/SPRI WD-1 “Wind Design Standard for Roofing Assemblies”.
C. ASTM International (ASTM):
    1. ASTM C 1289 – Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board.
    2. ASTM D 412 – Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers-Tension.
    3. ASTM D 624 – Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers.
    4. ASTM D 816 – Standard Test Methods for Rubber Cements.
    5. ASTM D 4637 – Standard Specification for EPDM Sheet Used In Single-Ply Roof Membrane.
    6. ASTM E 96 – Standard Test Methods for Water Vapor Transmission of Materials.
D. International Code Council (ICC):
    1. International Building Code (IBC).
E. National Roofing Contractors Association (NRCA) – Low Slope Roofing and Waterproofing Manual, Current Edition.
F. Sheet Metal and Air Conditioning Contractors National Association, Inc. (SMACNA) – Architectural Sheet Metal Manual.
G. Underwriters Laboratories (UL):
    1. TGFU R1306 – “Roofing Systems and Materials Guide”

2. UL-790 – Standard Test Method for Fire Tests of Roof Coverings.

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1.04 DESIGN CRITERIA

A. Wind Uplift Performance:
   1. Roof system is designed to withstand wind uplift forces as calculated using the current revision of ASCE-7.
   2. Design Wind Speed - 115 ultimate wind speed.
   3. 90 MPH manufacturer wind speed warranty.

B. Fire Resistance Performance:
   1. Roof system will achieve a UL Class A rating when tested in accordance with UL-790.

C. Hail Rating: Severe Hail (SH)

D. Thermal Performance: Roof system will achieve a minimum R value not less than 30.

E. Drainage: Provide a roof system with positive drainage where all standing water dissipates within 48 hours after precipitation ends.

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      b. Include the manufacturer's name and product name for each component.
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      a. Clearly indicate the fastening pattern for each component, including type of fastener, size, and spacing.

b. Delineate wind zones on the roof plan and provide corresponding fastening requirements for each zone based on ASCE 7 calculations.

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C. Proceed with roofing work only when weather conditions are in compliance with the manufacturer's recommended limitations, and when conditions will permit the work to proceed in accordance with the manufacturer's requirements and recommendations.

D. Proceed with work so new roofing materials are not subject to construction traffic. When necessary, new roof sections shall be protected and inspected upon completion for possible damage.
E. Provide protection, such as ¾ inch thick plywood, for all roof areas exposed to traffic during construction. Plywood must be smooth and free of fasteners and splinters.
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G. New roofing shall be complete and weathertight at the end of the work day.
H. Contaminants such as grease, fats and oils shall not be allowed to come in direct contact with the roofing membrane.

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A. At project closeout, provide to the Owner or Owners Representative an executed copy of the manufacturer's Total-System warranty, outlining its terms, conditions, and exclusions from coverage.
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    4. Coverage to be extended to include 32 hours of accidental puncture warranty in accordance with the stated terms of the warranty document.
    5. Coverage to be extended to include roof edge metal water tightness in accordance with terms stated in the Warranty document.

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A. Acceptable Manufacturer:
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B. Base Flashing: Provide a waterproof, fully adhered base flashing system at all penetrations, plane transitions and terminations.

C. Insulation: Provide a roof insulation system beneath the finish membrane.

2.03 ETHYLENE, PROPYLENE, DIENE TERPOLYMER (EPDM) MEMBRANE

A. Sure-Seal Non-Reinforced Membrane: Cured, non-reinforced EPDM membrane meeting the requirements of ASTM D 4637 Type I.

1. Color: Black.
2. Attachment Method: Fully adhered.
3. Membrane Thickness: 90 mil, 0.090 inch, over 55 mil (0.050 inch) fleece; 145 mils total.
4. Sheet Dimensions:
    a. Width: 10 feet (3.05 m) maximum.
    b. Length: 100 feet (30.5 m) maximum.
4. Performance:
    a. Tensile Strength: 1550 psi (10.7 MPa) minimum.
    b. Tear Resistance: 200 lbf/in (35 Kn/m) minimum.
        c. Elongation: 480 percent.

2.04 COVERBOARDS

A. Cover Boards: High-density polyisocyanurate roof insulation board with coated glass-fiber facers, complying with ASTM C1289, Type II, Class 4, Grade 1.

1. **Manufacturers:** Carlisle or Approved equal.
2. **Product:** SecurShield HD Plus Cover Board.
3. Thickness: ½ inch.
4. Surface Finish: Coated glass-fiber facer.

2.05 INSULATION ADHESIVE

A. Flexible FAST Adhesive: A spray or extruded applied, two-component polyurethane, low-rise expanding foam adhesive used for attaching approved insulations to compatible substrates (concrete, cellular lightweight insulating concrete, gypsum, cementitious wood fiber, wood or steel) or existing smooth or gravel surfaced BUR, modified bitumen or cap sheets.

B. Flexible FAST Dual Cartridge Adhesive: A two-component, polyurethane construction grade, low-rise expanding adhesive designed for bonding insulation to various substrates using a portable applicator.

C. Flexible FAST Dual Tank Adhesive: A two-component, polyurethane construction grade, low-rise expanding adhesive designed for bonding insulation to various substrates using a portable applicator.

D. Flexible FAST 5 gallon Jug Adhesive: A two-component, polyurethane construction grade, low-rise expanding adhesive designed for bonding insulation to various substrates, packaged for use with spray application rigs.

E. OlyBond 500 BA - A two-component, polyurethane, low-rise expanding adhesive used to bond insulation to various substrates using a mechanical dispenser system.

F. OlyBond Spot Shot – A two-component, polyurethane construction grade, low-rise expanding adhesive designed for bonding insulation to various substrates using a portable applicator.

G. One-Step: A two-component, polyurethane construction grade, low-rise expanding adhesive designed for bonding insulation to various substrates using a portable applicator.

2.06 FLASHING ACCESSORIES

A. Sure-Seal (black)/Sure-White (white) Pressure-Sensitive Pipe Seals with Factory-Applied TAPE on the deck flange are available for use with Sure-Seal/Sure-White Roofing systems.

B. Sure-Seal/Sure-White Pressure-Sensitive (PS) Inside/Outside Corner: A 7 inch by 9 inch precut 60-mil thick Elastoform Flashing with a 30-mil Factory-Applied TAPE.

C. Sure-Seal/Sure-White Pressure-Sensitive Cured Cover Strip: Sure-Seal or Sure-White 60-mil cured EPDM membrane laminated to a nominal 35-mil cured Factory-Applied TAPE.

D. Sure-Seal/Sure-White Pressure-Sensitive “T” Joint Covers: A factory cut uncured 60-mil thick EPDM flashing laminated to a nominal 35-mil Factory-Applied TAPE, used to overlay field splice intersections and to cover field splices at angle changes. Available in 6 inch by 6 inch and 12 inch by 12 inch for Sure-Seal applications, and 6 inch by 6 inch for Sure-White applications.

E. Sure-Seal/Sure-White Pressure-Sensitive Elastoform Flashing: 60-mil thick uncured EPDM Flashing laminated to a 30-mil Factory-Applied Pressure-Sensitive TAPE used in conjunction with Sure-Seal Primer.

F. Sure-Seal Pressure-Sensitive RUSS (Reinforced Universal Securement Strip):

1. 6 inch (152 mm) RUSS: A nominal 6 inch (152 mm) wide, 45-mil thick reinforced EPDM membrane with a nominal 3 inch (76 mm) wide 30-mil thick cured synthetic rubber pressure-sensitive adhesive laminated to one edge. This product provides perimeter securement, and additional membrane securement at angle changes for Adhered, Ballasted, and Mechanically Fastened Roofing Systems.

2.08 CLEANERS, PRIMERS, ADHESIVES AND SEALANTS

A. Carlisle Weathered Membrane Cleaner: Clear, solvent-based cleaner used to loosen and remove contaminants from the surface of exposed EPDM membrane prior to applying EPDM Primer.

B. Sure-Seal SecurTAPE: 3 inch (76 mm) or 6 inch (152 mm) wide by 100 foot (30.5 M) long splice tape used for splicing adjoining sections of EPDM membrane.

C. Sure-Seal HP-250 Primer: A solvent-based primer used to prepare the surface of EPDM membrane for application of Splice Tape or Pressure-Sensitive products.

D. Sure-Seal/Sure-White Lap Sealant: A heavy-bodied material (trowel or gun-consistency) used to seal the exposed edges of a membrane splice.

1. Sure-Seal Lap Sealant: Black sealant for use with Sure-Seal (black) Roofing Systems.

2. Sure-White Lap Sealant: White sealant for use with Sure-White (white-on-black) Roofing Systems.

E. 90-8-30A Bonding Adhesive: A high-strength, yellow colored, synthetic rubber adhesive used for bonding Sure-Seal/Sure-White EPDM membranes to various surfaces.

F. Flexible FAST Adhesive: A spray or extruded applied, two-component, polyurethane, low-rise expanding foam adhesive used to securely bond FleeceBACK membranes to a variety of substrates.

G. Water Cut-Off Mastic: A one-component, low viscosity, self wetting, Butyl blend mastic used as a compression sealing agent between EPDM membranes and applicable substrates.

H. Universal Single-Ply Sealant: A 100 percent solids, solvent free, one-part, polyether sealant that provides a weather tight sealant to a variety of building substrates; used as a termination bar sealant. Available in white only.

2.08 EDGINGS AND TERMINATIONS

A. SecurEdge EX Snap-On Fascia: A snap-on edge system consisting of a 24 gauge galvanized metal water dam. Finish as noted on the Finish Schedule of the Contract Drawings.

2.09 FASTENING COMPONENTS

A. HP Fastener: Threaded, black epoxy electro-deposition coated (E-Coat) fastener for use with steel, wood plank or oriented strand board (OSB).

B. Seam Fastening Plate: 2 inch (51 mm) diameter metal plate for insulation, membrane and RUSS attachment.

C. Insulation Fastening Plate: Nominal 3 inch (76 mm) diameter FM approved metal plate used for insulation attachment.

PART 3 EXECUTION

3.01 EXAMINATION

A. Do not begin installation until substrates have been properly prepared.

B. If substrate preparation is the responsibility of another installer, notify the Consultant of unsatisfactory preparation before proceeding.

3.02 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

C. Do not commence work until all other work trades have completed jobs that require them to traverse the deck on foot or with equipment.

3.03 INSULATION – SYSTEM DESIGN

A. Top Layer:

1. Type:Secureshield HD Plus
2. Thickness: ½ inches.
3. Attachment Method: Mechanically Fasten

3.04 INSULATION PLACEMENT

A. Install insulation or membrane underlayment in multiple layers over the substrate with boards butted tightly together with no joints or gaps greater than ¼ inch (6 mm). Stagger joints both horizontally and vertically.

B. Secure insulation to the substrate with the required mechanical fasteners or insulation adhesive in accordance with the manufacturer's current application guidelines.

C. Do not install wet, damaged or warped insulation boards.

D. Stagger joints in one direction unless joints are to be taped. Install insulation boards snug. Gaps between board joints shall not exceed ¼ inch (6 mm). Fill all gaps in excess of ¼ inch (6 mm) with same insulation material.

E. Wood nailers must be at least 3 ½ inches (89 mm) wide or 1 inch (25 mm) wider than adjacent metal flanges. Thickness must equal that of insulation but not less than 1 inch (25 mm) thickness.

F. Miter and fill the edges of the insulation boards at ridges, valleys and other changes in plane to prevent open joints or irregular surfaces. Avoid breaking or crushing the insulation at the corners.

G. Do not install any more insulation than will be completely waterproofed each day.

3.05 INSULATION ATTACHMENT

A. Adhesive coverage rate or fastening pattern as necessary to achieve the specified attachment and uplift rating. For adhesive, press each board firmly into place after adhesive develops strings when touched, typically 1-1/2 to 2 minutes after adhesive was applied, and roll with a weighted roller. Add temporary weight and use relief cuts to ensure boards are well adhered. Stagger the joints of additional layers by a minimum of 6 inches (152 mm).

3.06 MEMBRANE PLACEMENT AND ATTACHMENT (Fully Adhered)

A. Position and unroll successive sheets and align to provide for a minimum 3 inch (76 mm) wide splice.

B. Fold adjacent sheets in half lengthwise to expose an approximate 10 foot (3046 mm) wide substrate area.

C. Membrane which will have the adjacent sheet spliced over it should be adhered to the substrate first. In this fashion, selvage edge splice area will not be contaminated by setting splice edge into the FAST or Flexible FAST Adhesive.

D. Spray or extrude FAST or Flexible FAST Adhesive onto the substrate and allow to foam up approximately 1/8 inch (3 mm). Wait for the adhesive to achieve "string" when a small object is lifted out if the adhesive.

E. Place the membrane into adhesive after adhesive develops strings when touched, typically 1-1/2 to 2 minutes after adhesive was applied, and roll with a 150 lb (68 kg) segmented weighted roller.
F. Apply FAST Adhesive to the substrate and continue process described above until all sheets are fully bonded, allowing for necessary splice overlaps at selvage edges. At end laps (along the width of the sheet) membrane shall be butted together which will be overlaid with 6 inch wide Pressure-Sensitive Cured Cover Strip or Pressure-Sensitive Overlayment Strip.

3.07 MEMBRANE SPLICING (Tape Splice – Factory Applied)

A. Overlap adjacent sheets and mark a line ½ inch out from the top sheet.
B. Fold the top sheet back and clean the dry splice area (minimum 2 ½ inches (64 mm wide) of both membrane sheets with Sure-Seal Primer as required by the membrane manufacturer.
C. Where Splice Tape is not Factory-Applied, apply Splice Tape to bottom sheet with the edge of the release film along the marked line. Press tape onto the sheet using hand pressure. Overlap tape roll ends a minimum of 1 inch (13 mm).
D. Remove the release film and press the top sheet onto the tape using hand pressure.
E. Roll the seam toward the splice edge with a 2 inch (51 mm) wide steel roller.
F. Install Pressure-Sensitive “T” Joint Cover, a 6 inch wide (152 mm) section of Pressure-Sensitive Elastoform Flashing over all field splice intersections.
G. When using non-Pressure-Sensitive Elastoform Flashing or Elastoform Flashing, seal edges of flashing with Lap Sealant.
H. The use of Lap Sealant with tape splices is optional except at tape overlaps and cut edges of reinforced membrane where Lap Sealant is required.

3.08 FLASHING

A. Wall and curb flashing shall be cured EPDM membrane. Continue the deck membrane as wall flashing where practicable.
B. Follow manufacturer’s typical flashing procedures for all wall, curb, and penetration flashing including metal edging/coping and roof drain applications.

3.09 WALKWAYS

A. Install walkways at all traffic concentration points (such as roof hatches, access doors, rooftop ladders, etc.) and all locations as identified on the Contract Drawings.
B. Adhere walkways pads to the EPDM membrane in accordance with the manufacturer’s current application guidelines.

3.10 DAILY SEALS

A. On phased roofing, when the completion of flashings and terminations is not achieved by the end of the work day, a daily seal must be performed to temporarily close the membrane to prevent water infiltration.
B. Use Sure-Seal Pourable Sealer or other acceptable membrane seal in accordance with the manufacturer’s requirements.

SECTION 07 53 00

EPDM THERMOSET SINGLE-PLY ROOFING - *BASE BID*

PART 1 GENERAL

1.01 SECTION INCLUDES

A. EPDM thermoset single-ply roofing
B. Membrane flashings.
C. Metal flashings.
D. Roof insulation.

1.02 RELATED SECTIONS

A. Section 05 31 23 – Steel Roof Decking.
B. Section 06 10 00 – Rough Carpentry.
C. Section 07 62 00 – Sheet Metal Flashing and Trim.
D. Section 07 70 00 – Roof and Wall Specialties and Accessories.

1.03 REFERENCES

A. American Society of Civil Engineers (ASCE) – ASCE 7 - Minimum Design Loads for Buildings and Other Structures, Current Revision.
B. ANSI/SPRI WD-1 “Wind Design Standard for Roofing Assemblies”.
C. ASTM International (ASTM):
    1. ASTM C 1289 – Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board.
    2. ASTM D 412 – Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers-Tension.
    3. ASTM D 624 – Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers.
    4. ASTM D 816 – Standard Test Methods for Rubber Cements.
    5. ASTM D 4637 – Standard Specification for EPDM Sheet Used In Single-Ply Roof Membrane.
    6. ASTM E 96 – Standard Test Methods for Water Vapor Transmission of Materials.
D. International Code Council (ICC):
    1. International Building Code (IBC).
E. National Roofing Contractors Association (NRCA) – Low Slope Roofing and Waterproofing Manual, Current Edition.
F. Sheet Metal and Air Conditioning Contractors National Association, Inc. (SMACNA) – Architectural Sheet Metal Manual.
G. Underwriters Laboratories (UL):
    1. TGFU R1306 – “Roofing Systems and Materials Guide”

2. UL-790 – Standard Test Method for Fire Tests of Roof Coverings.

H. ANSI/ASHRAE/IESNA Standard 90.1 (2007): Energy Standard for Buildings Except Low-Rise Residential Buildings

1.04 DESIGN CRITERIA

A. Wind Uplift Performance:
   1. Roof system is designed to withstand wind uplift forces as calculated using the current revision of ASCE-7.
   2. Design Wind Speed - 115 ultimate wind speed.
   3. 72 MPH manufacturer wind speed warranty.

B. Fire Resistance Performance:
   1. Roof system will achieve a UL Class A rating when tested in accordance with UL-790.

C. Hail Rating: Severe Hail (SH)

D. Thermal Performance: Roof system will achieve a minimum R value not less than 30.

E. Drainage: Provide a roof system with positive drainage where all standing water dissipates within 48 hours after precipitation ends.

F. Building Codes:
   1. Roof system will meet the requirements of all federal, state and local code bodies having jurisdiction.

1.05 SUBMITTALS

A. Submit under provisions of Section 01 30 00.

B. Product Data: Manufacturer's data sheets on each product to be used, including:
   1. Preparation instructions and recommendations.
   2. Storage and handling requirements and recommendations.
   3. Installation methods.

C. Detail Drawings:
   1. Submit approved plan, section, elevation or isometric drawings which detail the appropriate methods for all flashing conditions found on the project.
   2. Coordinate approved drawings with locations found on the Contract Drawings.

D. Roof Assembly Letter: Provide a roof assembly letter on [Roof Manufacturer's] letterhead for each roof area, detailing the following:
   1. System Components:
      a. List all components of the roofing assembly in the order of installation, from the structural deck up to the membrane.
      b. Include the manufacturer's name and product name for each component.
   2. Fastening Patterns and Zones:
      a. Clearly indicate the fastening pattern for each component, including type of fastener, size, and spacing.

b. Delineate wind zones on the roof plan and provide corresponding fastening requirements for each zone based on ASCE 7 calculations.

3. Warranty Requirements:
   a. State the specific installation requirements to achieve the full manufacturer's warranty.
   b. Include any limitations or exclusions to the warranty.

E. Selection Samples: For each finish product specified, two complete sets of chips representing the manufacturer's full range of available colors, membranes, and thicknesses.

F. Verification Samples: For each finish product specified, two samples, minimum size 4 inches (100 mm) square representing actual product, color, and patterns.

1.06 QUALITY ASSURANCE

A. Manufacturer Qualifications: All products specified in this section will be supplied by a single manufacturer with a minimum of twenty (20) years experience.

B. Installer Qualifications:
   1. All products listed in this section are to be installed by a single installer with a minimum of five (5) years demonstrated experience in installing products of the same type and scope as specified.
   2. Installer must be capable of extending the Manufacturer's No Dollar Limit guarantee.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Store products in manufacturer's unopened packaging until ready for installation.

B. Store and dispose of hazardous materials, and materials contaminated by hazardous materials, in accordance with requirements of local authorities having jurisdiction.

C. Safety Data Sheets (SDS) must be on location at all times during the transportation, storage and application of materials.

D. When loading materials onto the roof, the Roofing Applicator must comply with the requirements of the building owner to prevent overloading and possible disturbance to the building structure.

1.08 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by the manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

B. When positioning membrane sheets, exercise care to locate all field splices away from low spots and out of drain sumps. All field splices should be shingled to prevent bucking of water.

C. Proceed with roofing work only when weather conditions are in compliance with the manufacturer's recommended limitations, and when conditions will permit the work to proceed in accordance with the manufacturer's requirements and recommendations.

D. Proceed with work so new roofing materials are not subject to construction traffic. When necessary, new roof sections shall be protected and inspected upon completion for possible damage.
E. Provide protection, such as ¾ inch thick plywood, for all roof areas exposed to traffic during construction. Plywood must be smooth and free of fasteners and splinters.
F. The surface on which the insulation or roofing membrane is to be applied shall be clean, smooth, dry, and free of projections or contaminants that would prevent proper application of or be incompatible with the new installation, such as fins, sharp edges, foreign materials, oil and grease.
G. New roofing shall be complete and weathertight at the end of the work day.
H. Contaminants such as grease, fats and oils shall not be allowed to come in direct contact with the roofing membrane.

1.09 WARRANTY

A. At project closeout, provide to the Owner or Owners Representative an executed copy of the manufacturer's Total-System warranty, outlining its terms, conditions, and exclusions from coverage.
    1. 30 years Total-System warranty, outlining its terms, conditions, and exclusions from coverage.
    2. Coverage to be extended to include damage caused by a maximum 72 mph wind speeds in accordance with terms stated in the Warranty document.
    3. Coverage to be extended to include damage caused by a maximum 2 inch diameter hail in accordance with terms stated in the Warranty document.
    4. Coverage to be extended to include 32 hours of accidental puncture warranty in accordance with the stated terms of the warranty document.
    5. Coverage to be extended to include roof edge metal water tightness in accordance with terms stated in the Warranty document.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Acceptable Manufacturer:
    1. Carlisle SynTec Systems (Basis of Design)
    2. Elevate Roofing Systems
    3. Johns Manville Roofing Systems
B. Requests for substitutions will not be considered.

2.02 SCOPE / APPLICATION

A. Roof System: Provide a waterproof roof system, capable of withstanding uplift forces as specified in Design Criteria.

B. Base Flashing: Provide a waterproof, fully adhered base flashing system at all penetrations, plane transitions and terminations.
C. Insulation: Provide a roof insulation system beneath the finish membrane.

2.03 ETHYLENE, PROPYLENE, DIENE TERPOLYMER (EPDM) MEMBRANE

A. Sure-Seal Non-Reinforced Membrane: Cured, non-reinforced EPDM membrane meeting the requirements of ASTM D 4637 Type I.
    1. Color: Black.
    2. Attachment Method: BallastedFully adhered.
    3. Membrane Thickness: 90 mil nominal.
    4. Sheet Dimensions:
        a. Width: 10 feet (3.05 m) maximum.
        b. Length: 100 feet (30.5 m) maximum.
    4. Performance:
        a. Tensile Strength: 1550 psi (10.7 MPa) minimum.
        b. Tear Resistance: 200 lbf/in (35 Kn/m) minimum.
            c. Elongation: 480 percent.

2.04 INSULATION

A. Polyisocyanurate InsulBase: Carlisle InsulBase. Rigid board with glass fiber reinforced facers (GRF) on both sides, meeting or exceeding the requirements of ASTM C 1289, Type II, Class 1.
    1. Compressive Strength: Grade 2 (20 psi) (138 kPa).
    2. Density: 2 lb per cubic foot (24 kg/cu m) minimum.
    3. Size: 1” thick

2.05 FLASHING ACCESSORIES

A. Sure-Seal (black)/Sure-White (white) Pressure-Sensitive Pipe Seals with Factory-Applied TAPE on the deck flange are available for use with Sure-Seal/Sure-White Roofing systems.
B. Sure-Seal/Sure-White Pressure-Sensitive (PS) Inside/Outside Corner: A 7 inch by 9 inch precut 60-mil thick Elastoform Flashing with a 30-mil Factory-Applied TAPE.
C. Sure-Seal/Sure-White Pressure-Sensitive Cured Cover Strip: Sure-Seal or Sure-White 60-mil cured EPDM membrane laminated to a nominal 35-mil cured Factory-Applied TAPE.
D. Sure-Seal/Sure-White Pressure-Sensitive “T” Joint Covers: A factory cut uncured 60-mil thick EPDM flashing laminated to a nominal 35-mil Factory-Applied TAPE, used to overlay field splice intersections and to cover field splices at angle changes. Available in 6 inch by 6 inch and 12 inch by 12 inch for Sure-Seal applications, and 6 inch by 6 inch for Sure-White applications.
E. Sure-Seal/Sure-White Pressure-Sensitive Elastoform Flashing: 60-mil thick uncured EPDM Flashing laminated to a 30-mil Factory-Applied Pressure-Sensitive TAPE used in conjunction with Sure-Seal Primer.
F. Sure-Seal Pressure-Sensitive RUSS (Reinforced Universal Securement Strip):

1. 6 inch (152 mm) RUSS: A nominal 6 inch (152 mm) wide, 45-mil thick reinforced EPDM membrane with a nominal 3 inch (76 mm) wide 30-mil thick cured synthetic rubber pressure-sensitive adhesive laminated to one edge. This product provides perimeter securement, and additional membrane securement at angle changes for Adhered, Ballasted, and Mechanically Fastened Roofing Systems.

2.06 CLEANERS, PRIMERS, ADHESIVES AND SEALANTS

A. Carlisle Weathered Membrane Cleaner: Clear, solvent-based cleaner used to loosen and remove contaminants from the surface of exposed EPDM membrane prior to applying EPDM Primer.

B. Sure-Seal SecurTAPE: 3 inch (76 mm) or 6 inch (152 mm) wide by 100 foot (30.5 M) long splice tape used for splicing adjoining sections of EPDM membrane.

C. Sure-Seal HP-250 Primer: A solvent-based primer used to prepare the surface of EPDM membrane for application of Splice Tape or Pressure-Sensitive products.

D. Sure-Seal/Sure-White Lap Sealant: A heavy-bodied material (trowel or gun-consistency) used to seal the exposed edges of a membrane splice.
   1. Sure-Seal Lap Sealant: Black sealant for use with Sure-Seal (black) Roofing Systems.
   2. Sure-White Lap Sealant: White sealant for use with Sure-White (white-on-black) Roofing Systems.

E. 90-8-30A Bonding Adhesive: A high-strength, yellow colored, synthetic rubber adhesive used for bonding Sure-Seal/Sure-White EPDM membranes to various surfaces.

F. Flexible FAST Adhesive: A spray or extruded applied, two-component, polyurethane, low-rise expanding foam adhesive used to securely bond FleeceBACK membranes to a variety of substrates.

G. Water Cut-Off Mastic: A one-component, low viscosity, self wetting, Butyl blend mastic used as a compression sealing agent between EPDM membranes and applicable substrates.

H. Universal Single-Ply Sealant: A 100 percent solids, solvent free, one-part, polyether sealant that provides a weather tight sealant to a variety of building substrates; used as a termination bar sealant. Available in white only.

2.07 EDGINGS AND TERMINATIONS

A. SecurEdge EX Snap-On Fascia: A snap-on edge system consisting of a 24 gauge galvanized metal water dam. Finish as noted on the Finish Schedule of the Contract Drawings.

2.8 FASTENING COMPONENTS

A. HP Fastener: Threaded, black epoxy electro-deposition coated (E-Coat) fastener for use with steel, wood plank or oriented strand board (OSB).

B. Seam Fastening Plate: 2 inch (51 mm) diameter metal plate for insulation, membrane and RUSS attachment.

C. Insulation Fastening Plate: Nominal 3 inch (76 mm) diameter FM approved metal plate used for insulation attachment.

2.9 PROTECTION MAT

A. A protective mat/cushion layer shall be installed directly over the 90-mil EPDM membrane prior to ballast placement to prevent puncture and mechanical damage. BALLAST

2.10 BALLAST

B. 90-mil EPDM membrane loose laid and held in place with 1-1/2" nominal rounded water worn gravel conforming to gradation #4 in accordance with the ASTM-D448 method of sizing. Coverage rate shall be a minimum of 12 pounds per square foot in the field and 2-1/2” nominal conforming to graduation #1 or #2 in zone 3 at a coverage rate of 15 pounds per square foot in minimum 12’ x 12’ corners.

2.11 CONCRETE ROOF WALKWAY PAVER

A. Furnish and install Westile High-Density Concrete Roof Pavers (24" x 24" x 2" nominal, minimum 8,000 PSI) featuring chamfered edges and a standard slip-resistant finish, placed directly over a manufacturer-approved 60mil EPDM membrane protection pad to establish a designated maintenance walkway.

PART 3 EXECUTION

3.01 EXAMINATION

A. Do not begin installation until substrates have been properly prepared.

B. If substrate preparation is the responsibility of another installer, notify the Consultant of unsatisfactory preparation before proceeding.

3.02 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

C. Do not commence work until all other work trades have completed jobs that require them to traverse the deck on foot or with equipment.

3.025 BALLAST REMOVAL AND PREPARATION

A. Equipment Construction (Box Screen): The Contractor shall clean the existing rock ballast utilizing a site-built or manufactured "Box Screen." This apparatus must consist of a rigid framed box with heavy-duty wire mesh stretched securely across the top or middle, positioned over a debris catch basin or containment area.

B. B. Execution & Sifting Process: Workers shall shovel the existing dirty ballast directly onto the wire mesh and manually agitate the material to ensure all dirt, pine needles, and broken, sharp rock fragments (fines) fall through the mesh into the catch below.

C. C. Quality Control & Debris Removal: Only smooth, clean, full-sized ballast rock retained on top of the mesh shall be approved for reinstallation over the new roof assembly. The sifted waste material captured in the lower box must be regularly emptied, bagged, and legally disposed of

off-site.

D. D. Membrane & Deck Protection: To prevent puncture damage or crushing of the insulation during the shoveling and sifting process, the box screen setup must be placed over protective plywood or heavy-duty protection mats at all times while in operation.

3.03 INSULATION – SYSTEM DESIGN

A. Base Layers:
   1. Type: Polyisocyanurate 20 PSI – ASTM C 1289, Type II, Class 1.
   2. Thickness: 1”.
   3. Attachment Method: Clean & blow off existing insulation - Loose Laid

3.04 INSULATION PLACEMENT

A. Clean off existing insulation to remain.
B. Install insulation or membrane underlayment in multiple layers over the substrate with boards butted tightly together with no joints or gaps greater than ¼ inch (6 mm). Stagger joints both horizontally and vertically.
C. Secure insulation to the substrate with the required mechanical fasteners or insulation adhesive in accordance with the manufacturer’s current application guidelines.
D. Do not install wet, damaged or warped insulation boards.
E. Stagger joints in one direction unless joints are to be taped. Install insulation boards snug. Gaps between board joints shall not exceed ¼ inch (6 mm). Fill all gaps in excess of ¼ inch (6 mm) with same insulation material.
F. Wood nailers must be at least 3 ½ inches (89 mm) wide or 1 inch (25 mm) wider than adjacent metal flanges. Thickness must equal that of insulation but not less than 1 inch (25 mm) thickness.
G. Miter and fill the edges of the insulation boards at ridges, valleys and other changes in plane to prevent open joints or irregular surfaces. Avoid breaking or crushing the insulation at the corners.
H. Do not install any more insulation than will be completely waterproofed each day.

3.05 INSULATION ATTACHMENT

A. Attachment: Insulation boards shall be laid loose without mechanical fasteners or adhesive, relying on the overlying ballasted system assembly for securement.

3.06 MEMBRANE PLACEMENT AND ATTACHMENT (Ballasted System)

A. Position and unroll membrane sheets over the prepared substrate without stretching. Allow sheets to relax unconstrained for a minimum of 30 minutes prior to splicing or securing.
B. Align adjacent sheets to provide for a minimum 3-inch (76 mm) side lap and 6-inch (152 mm) end lap, or per manufacturer guidelines for ballast-secured splice details.
C. Membrane sheets shall be laid loose over the substrate without adhesive application, except at perimeters, penetrations, wall flashings, and angle changes where mechanical attachment or full adhesion is required per specification and manufacturer details.

D. Complete all field splices, end laps, and perimeter securement in accordance with manufacturer technical specifications prior to the placement of protective layers or ballast.
E. Protection Mat Installation: Unroll and lay a non-woven polypropylene geotextile protection mat over the completed, inspected membrane prior to ballast placement. Overlap all adjacent protection mat sheets a minimum of 6 inches (152 mm) to ensure continuous protection against ballast abrasion and puncture.

3.07 MEMBRANE SPLICING (Tape Splice – Factory Applied)

A. Overlap adjacent sheets and mark a line ½ inch out from the top sheet.
B. Fold the top sheet back and clean the dry splice area (minimum 2 ½ inches (64 mm wide) of both membrane sheets with Sure-Seal Primer as required by the membrane manufacturer.
C. Where Splice Tape is not Factory-Applied, apply Splice Tape to bottom sheet with the edge of the release film along the marked line. Press tape onto the sheet using hand pressure. Overlap tape roll ends a minimum of 1 inch (13 mm).
D. Remove the release film and press the top sheet onto the tape using hand pressure.
E. Roll the seam toward the splice edge with a 2 inch (51 mm) wide steel roller.
F. Install Pressure-Sensitive “T” Joint Cover, a 6 inch wide (152 mm) section of Pressure-Sensitive Elastoform Flashing over all field splice intersections.
G. When using non-Pressure-Sensitive Elastoform Flashing or Elastoform Flashing, seal edges of flashing with Lap Sealant.
H. The use of Lap Sealant with tape splices is optional except at tape overlaps and cut edges of reinforced membrane where Lap Sealant is required.

3.08 FLASHING

A. Wall and curb flashing shall be 90 MIL cured EPDM membrane. Continue the deck membrane as wall flashing where practicable.
B. Follow manufacturer’s typical flashing procedures for all wall, curb, and penetration flashing including metal edging/coping and roof drain applications.

3.09 WALKWAYS

A. Install walkways at all traffic concentration points (such as roof hatches, access doors, rooftop ladders, etc.) and all locations as identified on the Contract Drawings.
B. Non-ballasted sections - Adhere walkways pads to the EPDM membrane in accordance with the manufacturer’s current application guidelines.
C. Ballasted sections - loose lay 60mil EPDM membrane under new concrete paver in accordance with the manufacturer’s current application guidelines.

3.10 DAILY SEALS

A. On phased roofing, when the completion of flashings and terminations is not achieved by the end of the work day, a daily seal must be performed to temporarily close the membrane to prevent water infiltration.

B. Use Sure-Seal Pourable Sealer or other acceptable membrane seal in accordance with the manufacturer's requirements.

3.11 CLEAN UP

A. Perform daily clean-up to collect all wrappings, empty containers, paper, and other debris from the project site. Upon completion, all debris must be disposed of in a legally acceptable manner.

B. Prior to the manufacturer's inspection for warranty, the applicator must perform a pre-inspection to review all work and to verify all flashing has been completed as well as the application of all caulking.

3.12 PROTECTION

A. Protect installed products until completion of project.

B. Touch-up, repair or replace damaged products before Substantial Completion.

END OF SECTIONSECTION 07 53 00

SECTION 07 62 00
SHEET METAL FLASHING AND TRIM

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary
B. Conditions and Division 01 Specification Sections, apply to this Section.

PART 2 SUMMARY

2.01 SECTION INCLUDES:

A. Surface mounted counter flashing
B. Scuppers
C. Downspouts
D. Conductor heads

2.02 RELATED REQUIREMENTS:

A. Division 06 "Rough Carpentry" for wood nailers, curbs, and blocking.
B. Division 07 “Membrane Roofing” for materials and installation of sheet metal flashing and trim integral with roofing.

2.03 COORDINATION

A. Coordinate sheet metal flashing and trim layout and seams with sizes and locations of penetrations to be flashed, and joints and seams in adjacent materials.
B. Coordinate sheet metal flashing and trim installation with adjoining roofing and wall materials, joints, and seams to provide leak proof, secure, and noncorrosive installation.

2.04 PREINSTALLATION MEETINGS

A. Pre Installation Conference: Conduct conference at Project site.
    1. Review construction schedule. Verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
    2. Review special roof details, roof drainage, roof-penetration flashing, equipment curbs, and condition of other construction that affect sheet metal flashing and trim
    3. Review requirements for insurance and certificates if applicable.
    4. Review sheet metal flashing observation and repair procedures after flashing installation.

2.05 SUBMITTALS

A. Product Data: For each type of product.
    1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for each manufactured product and accessory.
    2. Shop Drawings: For sheet metal flashing and trim.
        a. Include plans, elevations, sections, and attachment details.
        b. Detail fabrication and installation layouts, expansion-joint locations, and keyed details. Distinguish between shop- and field-assembled work.

c. Include identification of material, thickness, weight, and finish for each item and location in Project.
d. Include details for forming, including profiles, shapes, seams, and dimensions.
e. Include details for joining, supporting, and securing, including layout and spacing of fasteners, cleats, clips, and other attachments. Include pattern of seams.
f. Include details of termination points and assemblies.
g. Include details of expansion joints and expansion-joint covers, including showing direction of expansion and contraction from fixed points.
h. Include details of roof-penetration flashing.
i. Include details of edge conditions, including eaves, ridges, valleys, rakes, crickets, and counter flashings as applicable.
j. Include details of special conditions.
k. Include details of connections to adjoining work.
l. Detail formed flashing and trim at scale of not less than 3 inches per 12 inches.

3. Samples for Initial Selection: For each type of sheet metal and accessory indicated with factory-applied finishes.
4. Samples for Verification: For each type of exposed finish.
   a. Sheet Metal Flashing: 12 inches (300 mm) long by actual width of unit, including finished seam and in required profile. Include fasteners, cleats, clips, closures, and other attachments.
   b. Trim, Metal Closures, Expansion Joints, Joint Intersections, and Miscellaneous Fabrications: 12 inches (300 mm) long and in required profile. Include fasteners and other exposed accessories.

2.06 INFORMATIONAL SUBMITTALS

A. Qualification Data: For fabricator.
B. Product Certificates: For each type of coping and roof edge flashing that is SPRI ES-1 tested.
C. Product Test Reports: For each product, for tests performed by a qualified testing agency.
D. Sample Warranty: For special warranty.

2.07 CLOSEOUT SUBMITTALS

A. Maintenance Data: For sheet metal flashing and trim, and its accessories, to include in maintenance manuals.

2.08 QUALITY ASSURANCE

A. Fabricator Qualifications: Employs skilled workers who custom fabricate sheet metal flashing and trim similar to that required for this Project and whose products have a record of successful in-service performance.
   1. For copings and roof edge flashings that are SPRI ES-1 tested, shop shall be listed as able to fabricate required details as tested and approved.

2.09 DELIVERY, STORAGE, AND HANDLING

A. Do not store sheet metal flashing and trim materials in contact with other materials that might cause staining, denting, or other surface damage. Store sheet metal flashing and trim materials away from uncured concrete and masonry.

B. Protect strippable protective covering on sheet metal flashing and trim from exposure to sunlight and high humidity, except to extent necessary for period of sheet metal flashing and trim installation.

2.10 WARRANTY

A. Special Warranty on Finishes: Manufacturer agrees to repair finish or replace sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within the specified warranty period.

1. Exposed Panel Finish: Deterioration includes, but is not limited to, the following:
    a. Color fading more than 5 Hunter units when tested according to ASTM D 2244.
    b. Chalking in excess of a No. 8 rating when tested according to ASTM D 4214.
    c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.
2. Finish Warranty Period: 20 years from date of Substantial Completion.

PART 3 PRODUCTS

3.01 PERFORMANCE REQUIREMENTS

A. General: Sheet metal flashing and trim assemblies shall withstand wind loads, structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim shall not rattle, leak, or loosen, and shall remain watertight.

B. Sheet Metal Standard for Flashing and Trim: Comply with NRCA's "The NRCA Roofing Manual" and SMACNA's "Architectural Sheet Metal Manual" requirements for dimensions and profiles shown unless more stringent requirements are indicated.

C. SPRI Wind Design Standard: Manufacture and install roof edge flashings tested according to SPRI ES-1 and capable of resisting the design pressure.

D. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes to prevent buckling, opening of joints, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

3.02 SHEET METALS

A. General: Protect mechanical and other finishes on exposed surfaces from damage by applying strippable, temporary protective film before shipping.

B. Metallic-Coated Steel Sheet: Provide aluminum-zinc alloy-coated steel sheet in accordance with ASTM A792/A792M, Class AZ50 coating designation, Grade 40; prepainted by coil-coating process to comply with ASTM A755/A755M.
    1. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.
    2. Thickness: 0.028 inch thick (24 gauge), unless indicated otherwise.
    3. Surface: Smooth, flat.
    4. Exposed Coil-Coated Finish:
        a. Two-Coat Fluoropolymer: AAMA 621. Fluoropolymer finish containing not less than 70 percent polyvinylidene fluoride (PVDF) resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
    5. Color: As selected Owner from manufacturer's full range.
    6. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester backer finish, consisting of prime coat and wash coat with minimum total dry film thickness of 0.5 mil.

C. TPO Coated Metal: A 24-gauge, hot-dipped galvanized steel sheet laminated with a layer of non-reinforced thermoplastic polyolefin (TPO) flashing. This product is designed to allow the direct heat-welding of TPO roof membranes to metal flashings, eliminating the need for stripping-in with adhesives.
    1. Material Standards:
        a. Steel Substrate: 24-gauge (0.6 mm) galvanized steel, G90 coating designation, conforming to ASTM A653.
        b. TPO Laminate: 0.035 inch (0.9 mm) nominal thickness, non-reinforced TPO membrane film.
        c. Total Thickness: Approximately 0.060 inch (1.5 mm).
    2. Physical Properties:
        a. Sheet Size: 4 feet by 10 feet (1.2m x 3.0m).
        b. Color: [White] [Gray] [Tan] [Select Special Color: Medium Bronze, Rock Brown, Terra Cotta, Slate Gray, Patina Green].
        c. Finish: Smooth, weldable surface.

3.03 MISCELLANEOUS MATERIALS

A. General: Provide materials and types of fasteners, solder, protective coatings, sealants, and other miscellaneous items as required for complete sheet metal flashing and trim installation and as recommended by manufacturer of primary sheet metal unless otherwise indicated.

B. Fasteners: Wood screws, annular threaded nails, self-tapping screws, self-locking rivets and bolts, and other suitable fasteners designed to withstand design loads and recommended by

manufacturer of primary sheet metal or manufactured item.

1. General: Blind fasteners or self-drilling screws, gasketed, with hex-washer head.
    a. Exposed Fasteners: Heads matching color of sheet metal using plastic caps or factory-applied coating. Provide metal-backed EPDM or PVC sealing washers under heads of exposed fasteners bearing on weather side of metal.
    b. Blind Fasteners: High-strength aluminum or stainless-steel rivets suitable for metal being fastened.
    c. Spikes and Ferrules: Same material as gutter; with spike with ferrule matching internal gutter width.
2. Fasteners for Zinc-Tin Alloy-Coated Stainless-Steel Sheet: Series 300 stainless steel.

C. Sealant Tape: Pressure-sensitive, 100 percent solids, polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape 1/2 inch wide and 1/8 inch thick.

D. Elastomeric Sealant: ASTM C 920, elastomeric polyurethane polymer sealant; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and trim and remain watertight.

3.04 FABRICATION, GENERAL

A. Custom fabricate sheet metal flashing and trim to comply with details indicated and recommendations in cited sheet metal standard that apply to design, dimensions, geometry, metal thickness, and other characteristics of item required.
    1. Fabricate sheet metal flashing and trim in shop to greatest extent possible.
    2. Fabricate sheet metal flashing and trim in thickness or weight needed to comply with performance requirements, but not less than that specified for each application and metal.
    3. Verify shapes and dimensions of surfaces to be covered and obtain field measurements for accurate fit before shop fabrication.
    4. Form sheet metal flashing and trim to fit substrates without excessive oil-canning, buckling, and tool marks; true to line, levels, and slopes; and with exposed edges folded back to form hems.
    5. Conceal fasteners and expansion provisions where possible. Do not use exposed fasteners on faces exposed to view.

B. Fabrication Tolerances: Fabricate sheet metal flashing and trim that is capable of installation to a tolerance of 1/4 inch in 20 feet on slope and location lines indicated on Drawings and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.

C. Expansion Provisions: Form metal for thermal expansion of exposed flashing and trim.
    1. Form expansion joints of intermeshing hooked flanges, not less than 1 inch deep, filled with butyl sealant concealed within joints.

2. Use lapped expansion joints only where indicated on Drawings.

D. Sealant Joints: Where movable, non expansion-type joints are required, form metal in accordance with cited sheet metal standard to provide for proper installation of elastomeric sealant.

E. Fabricate cleats and attachment devices from the same material as accessory being anchored or from compatible, noncorrosive metal.

F. Fabricate cleats and attachment devices of sizes as recommended by cited sheet metal standard for application, but not less than thickness of metal being secured.

G. Seams: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with elastomeric sealant unless otherwise recommended by sealant manufacturer for intended use. Rivet joints where necessary for strength.

3.06 WALL SHEET METAL FABRICATIONS

A. Manufactured Reglets and Counterflashing: Surface-mounted reglets of type and profile required, formed to provide secure interlocking of separate reglet and counterflashing pieces, and compatible with flashing indicated with factory-mitered and -welded corners and junctions.

1. Provide with slotted holes for fastening to substrate, with neoprene or other suitable weatherproofing washers, and with channel for sealant at top edge.
2. Material: Galvanized steel, not less than 0.028 inch thick (24 gauge).

B. Miscellaneous Sheet Metal Fabrications: Provide miscellaneous fascia and flashings as indicated on the Drawings.

1. Material: Metallic-coated sheet steel, coil-coat finished, thickness as indicated but not less than 0.028 inch thick (24 gauge).

PART 4 EXECUTION

4.01 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, substrate, and other conditions affecting performance of the Work.

1. Verify compliance with requirements for installation tolerances of substrates.
2. Verify that the substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.
3. Verify that air- or water-resistant barriers have been installed over sheathing or backing substrate to prevent air infiltration or water penetration.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

4.02 INSTALLATION, GENERAL

A. Install sheet metal flashing and trim to comply with details indicated and recommendations of cited sheet metal standard that apply to installation characteristics required unless otherwise indicated on Drawings.

1. Install fasteners, protective coatings, separators, sealants, and other miscellaneous items as required to complete sheet metal flashing and trim system.
2. Install sheet metal flashing and trim true to line, levels, and slopes. Provide uniform, neat seams with minimum exposure of sealant.
3. Anchor sheet metal flashing and trim and other components of the Work securely in place, with provisions for thermal and structural movement.
4. Install sheet metal flashing and trim to fit substrates and to result in watertight performance.
5. Install continuous cleats with fasteners spaced not more than 12 inches o.c.
6. Install exposed sheet metal flashing and trim free of oil-canning, buckling and tool marks.
7. Do not field cut sheet metal flashing and trim by torch.
8. Provide closure for sheet metal flashing terminations. The butt ends shall be fabricated with flanges that conform to the face of the adjacent wall or substrate.
9. Contractor shall round the corners of all flanges that are to be nailed and set over the completed membrane or flashings.
10. TPO Clad Metal: Butt joints must be covered with 2-inch wide aluminum tape (bond breaker) and stripped in with a 6-inch wide heat-welded strip of non-reinforced TPO flashing.
11. TPO Clad Metal: Field membrane shall be heat-welded directly to the coated metal flange with a minimum 1.5-inch weld width.

B. Metal Protection: Where dissimilar metals contact each other, or where metal contacts pressure treated wood or other corrosive substrates, protect against galvanic action or corrosion by painting contact surfaces with bituminous coating or by other permanent separation as recommended by sheet metal manufacturer or cited sheet metal standard.
    1. Coat concealed side of uncoated-aluminum and stainless steel sheet metal flashing and trim with bituminous coating where flashing and trim contact wood, ferrous metal, or cementitious construction.
C. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim.
    1. Space movement joints at maximum of 12 feet with no joints within 24 inches of corner or intersection.
D. Fasteners: Use fastener sizes that penetrate wood blocking or sheathing not less than 1-1/4 inches for nails and not less than 3/4 inch for wood screws.
E. Conceal fasteners and expansion provisions where possible in exposed work and locate to minimize possibility of leakage. Cover and seal fasteners and anchors as required for a tight installation.
F. Seal joints as required for watertight construction.
    1. Use sealant-filled joints unless otherwise indicated.

a. Form joints to completely conceal sealant.
b. When ambient temperature at time of installation is between 40 and 70 deg F, set joint members for 50 percent movement each way.
c. Adjust setting proportionately for installation at higher ambient temperatures.

2. Do not install sealant-type joints at temperatures below 40 deg F.
3. Prepare joints and apply sealants to comply with requirements in Section 07 92 00 "Joint Sealants."

4.05 ERECTION TOLERANCES

A. Installation Tolerances: Shim and align sheet metal flashing and trim within installed tolerance of 1/4 inch in 20 feet on slope and location lines indicated on Drawings and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.
B. Installation Tolerances: Shim and align sheet metal flashing and trim within installed tolerances specified in MCA's "Guide Specification for Residential Metal Roofing."

4.06 INSTALLATION TOLERANCES

A. Installation Tolerances: Shim and align sheet metal flashing and trim within installed tolerance of 1/4 inch in 20 feet on slope and location lines indicated on Drawings and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.

4.07 CLEANING

A. Clean exposed metal surfaces of substances that interfere with uniform oxidation and weathering.
B. Clean off excess sealants.

4.08 PROTECTION

A. Remove temporary protective coverings and strippable films as sheet metal flashing and trim are installed unless otherwise indicated in manufacturer's written installation instructions.
B. Maintain sheet metal flashing and trim in clean condition during construction.
C. Replace sheet metal flashing and trim that have been damaged or that have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

END OF SECTION 07 62 00

SECTION 07 71 00
ROOF SPECIALTIES

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Manufactured roof specialties, including fascias.

1.02 REFERENCE STANDARDS

A. AAMA 2605 - Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels (with Coil Coating Appendix).

B. ANSI/SPRI/FM 4435/ES-1 - Test Standard for Edge Systems Used with Low Slope Roofing Systems.

C. NRCA (RM) - The NRCA Roofing Manual.

1.03 SUBMITTALS

A. See Section 013000 - Administrative Requirements for submittal procedures.

B. Product Data: Provide data on shape of components, materials and finishes, anchor types and locations.

C. Shop Drawings: Indicate configuration and dimension of components, adjacent construction,required clearances and tolerances, and other affected work.

D. Samples: Submit two edge metal samples, 4 by 6 inches, illustrating component shape, finish,and color.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Roof Edge Flashings and Copings:

a. BASIS OF DESIGN: Carlisle: www.carlislesyntec.com

2.02 COMPONENTS

A. Roof Edge Flashings: Factory fabricated to sizes required; corners mitered; concealed fasteners.

a. Configuration: Fascia , and edge securement for roof membrane.

b. Pull-Off Resistance: Tested in accordance with ANSI/SPRI/FM 4435/ES-1 using test methods RE-1 and RE-2 to positive and negative design wind pressure as defined by applicable local building code.

c. Exposed Face Height: As indicated on drawings.

d. Material: Formed aluminum sheet, 0.050 inch thick, minimum.

e. Finish: 70 percent polyvinylidene fluoride (PVDF).

f. Color: TBD by owner.

g. Products:

i. **Carlisle SecurEdge EX Snap-On Fascia**

ii. Carlisle SecurEdge Drip Edge

2.03 FINISHES

A. PVDF (Polyvinylidene Fluoride) Coating: Superior Performance Organic Finish, AAMA 2605; multiple coat, thermally cured fluoropolymer finish system; color as selected from manufacturer's standard colors

2.04 ACCESSORIES

A. Sealant for Joints in Linear Components: As recommended by the component manufacturer.
B. Adhesive for Anchoring to Roof Membrane: Compatible with roof membrane and approved by roof membrane manufacturer.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that the deck, curbs, roof membrane, base flashing, and other items affecting work of this section are in place and positioned correctly.
B. See Section 077200 for information on roofing related accessories.

3.02 INSTALLATION

A. Install components in accordance with manufacturer's instructions and NRCA (RM) applicable requirements.
B. Seal joints within components when required by component manufacturer.
C. Anchor components securely.
D. Coordinate installation of components of this section with installation of roofing membrane and base flashings.
E. Coordinate installation of sealants and roofing cement with work of this section to ensure watertightness.

END OF SECTION 07 71 00

SECTION 07 72 30
ROOFTOP PIPE SUPPORTS

PART 1 - GENERAL

1.01 SUMMARY

A. This Section specifies the furnishing and installation of prefabricated, non-penetrating rooftop equipment support blocks for gas lines.

1.02 QUALITY ASSURANCE

A. A. Manufacturer's Instructions: Installation shall strictly comply with the manufacturer's current written instructions.

B. Load Capacity: Supports shall be installed to accommodate the weight of the gas line and all associated piping, while maintaining the specified distribution of load on the roof membrane.

PART 2 - PRODUCTS

2.01 ROOFTOP EQUIPMENT SUPPORT

A. Basis of Design Product: MIRO - 6 in. Rooftop Support Block with Strut - adjustable height to match gas line height.

   a. Material: 1. Block: High-density, recycled rubber or composite material resistant to UV, temperature extremes, and common roof chemicals.

   b. Strut: Integrated, hot-dipped galvanized or zinc-plated steel strut channel (1-5/8 inch series) for securing piping clamps.

   c. Accessories: Provide all necessary clamps, hardware, and fasteners required for securing the gas line to the support channel.

2.02 MEMBRANE PROTECTION

A. Protection Pad: Ensure the support block rests directly on a compatible protection pad or membrane slip sheet (minimum 60-mil non-woven fabric or rubber pad) to prevent membrane abrasion. *Note: If a protection pad is integrated into the block, no additional slip sheet is required.*

PART 3 - EXECUTION

3.01 INSTALLATION

A. Substrate Examination: Examine the roof surface to ensure it is clean, dry, and free of debris. Verify that the substrate is sound and properly prepared before placing supports.

B. Placement: Place supports in straight lines and level planes as required by the Drawings. Supports shall be located to distribute the load evenly and avoid roof seams or critical drainage paths.

C. Securing Piping: Secure the gas line firmly to the integrated strut channel using manufacturer-recommended clamps and hardware.

D. Non-Penetration: Ensure the installation is non-penetrating to the roof membrane.

END OF SECTION

SECTION 09 91 00
EXTERIOR PAINTING

PART 1 GENERAL

1.1 SUMMARY

A. Section includes surface preparation and application of protective enamel coating system for exterior exposed steel gas lines on the commercial roof.

1.02 QUALITY ASSURANCE

A. Manufacturer's Instructions: Comply strictly with Rust-Oleum Corporation's current written instructions and recommendations for surface preparation, primer, and finish coat application, curing time, and temperature/humidity conditions.

B. Installer Qualifications: Use only personnel trained and experienced in applying specified coating systems to metal substrates.

PART 2 - PRODUCTS

2.01 MATERIALS

A. Manufacturer: Rust-Oleum Corporation, or pre-approved equal.

B. Coating System: Rust-Oleum Protective Enamel (Oil-Based, Alkyd Enamel).

C. Primer: Rust-Oleum Clean Metal Primer (or Rust-Oleum Rusty Metal Primer, if applicable) for steel, compatible with the finish coat.

D. Finish Coat: Rust-Oleum Protective Enamel.

a. Color: [Safety Yellow]

b. Sheen: [Flat]

c. Thinners: Use only thinners specifically recommended by the coating manufacturer.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine surfaces to be coated. Verify that the gas line is clean, dry, and prepared according to manufacturer's recommendations. Proceed only after unsatisfactory conditions have been corrected.

3.02 SURFACE PREPARATION

A. General: Remove all dirt, grease, oil, loose rust, mill scale, and deteriorated coatings.

B. New Steel: Clean surface to near-white metal (SSPC-SP 10) for maximum coating adhesion. Ensure surfaces are fully dry prior to primer application.

C. Existing Rust (if applicable): Use power tool cleaning (SSPC-SP 3) to remove loose rust and scale before applying the recommended primer.

3.03 APPLICATION

A. Primer Application: Apply one coat of specified primer to achieve the manufacturer's

recommended dry film thickness (DFT).

B. Finish Coat Application: Apply two coats of Rust-Oleum Protective Enamel. 1. Allow primer and first coat to cure for the manufacturer's recommended duration before applying subsequent coats. 2. Apply coats uniformly by [Specify Method, e.g., brush, roller, or spray].

C. Total Dry Film Thickness (DFT): Achieve manufacturer's minimum required total DFT for severe exterior exposure.

3.04 CLEANING AND PROTECTION

A. A. Protect adjacent surfaces (roof membrane, equipment, vents) from overspray and spillage.

B. Clean spills and overspray immediately using proper solvents.

END OF SECTION